# BINOMIAL THEOREM

- **Binomial Expression**: An algebraic expression consisting of two terms with +e or –ve sign between them is called as binomial expression.
- **Binomial Theorem for Positive Integral Index**: If $n$ is a positive integer then,
  $$(x+a)^n = \sum_{r=0}^{n} {}^nC_r x^{n-r} a^r = {}^nC_0 x^n + {}^nC_1 x^{n-1} a^1 + {}^nC_2 x^{n-2} a^2 + \ldots + {}^nC_n a^n$$
- The number of terms in the expansion of $(x+a)^n$ is $n+1$.
- The sum of powers of $x$ and $a$ in any terms in the expansion of $(x+a)^n$ is $n$.
- The general term in the expansion of $(x+a)^n$ is $T_{r+1} = {}^nC_r x^{n-r} a^r$.
- ${}^nC_0, {}^nC_1, {}^nC_2, \ldots, {}^nC_n$ are binomial coefficients in the expansion of $(x+a)^n$.
- The binomial coefficients which equidistant from the beginning and end are equal. i.e. ${}^nC_0 = {}^nC_n$, ${}^nC_1 = {}^nC_{n-1}$, ${}^nC_2 = {}^nC_{n-2}, \ldots\ldots$
- In the expansion of $(x+a)^n$ r$^{th}$ term from the end is equal to (n – r + 2)$^{th}$ term from the beginning.
- $$(x-a)^n = \sum_{r=0}^{n} (-1)^r\, {}^nC_r x^{n-r} a^r = {}^nC_0 x^n - {}^nC_1 x^{n-1} a^1 + {}^nC_2 x^{n-2} a^2 - \ldots + (-1)^n\, {}^nC_n a^n$$
- $$(x+a)^n + (x-a)^n = \sum_{r=0}^{n} {}^nC_r x^{n-r} a^r + \sum_{r=0}^{n} (-1)^r\, {}^nC_r x^{n-r} a^r = 2\left({}^nC_0 x^n + {}^nC_2 x^{n-2} a^2 + {}^nC_4 x^{n-4} a^4 \ldots\right).$$
- $$(x+a)^n - (x-a)^n = \sum_{r=0}^{n} {}^nC_r x^{n-r} a^r - \sum_{r=0}^{n} (-1)^r\, {}^nC_r x^{n-r} a^r = 2\left({}^nC_1 x^{n-1} a^1 + {}^nC_3 x^{n-3} a^3 + {}^nC_5 x^{n-5} a^5 \ldots\right).$$
- The general term in the expansion of $(x-a)^n$ is $T_{r+1} = (-1)^r\, {}^nC_r x^{n-r} a^r$.
- The expansion of $(x+a)^n$ and $(a+x)^n$ are equal but their respective terms are not equal.
- $$(1+x)^n = \sum_{r=0}^{n} {}^nC_r x^r = {}^nC_0 + {}^nC_1 x^1 + {}^nC_2 x^2 + \ldots + {}^nC_n x^n$$
- The general term in the expansion of $(1+x)^n$ is $T_{r+1} = {}^nC_r x^r$.
- $$(1-x)^n = \sum_{r=0}^{n} (-1)^r\, {}^nC_r x^r = {}^nC_0 - {}^nC_1 x^1 + {}^nC_2 x^2 - \ldots + (-1)^n\, {}^nC_n x^n$$
- The general term in the expansion of $(1-x)^n$ is $T_{r+1} = (-1)^r\, {}^nC_r x^r$.
- The general term in the expansion of $(x_1 + x_2 + \ldots + x_p)^n$ is :
  $$T_{r+1} = \frac{(n_1 + n_2 + \ldots + n_p)!}{n_1! n_2! \ldots n_p!}\left(x_1^{n_1} x_2^{n_2} \ldots x_p^{n_p}\right)$$
- If $n$ is odd there will be two middle terms in the expansion of $(x+a)^n$ which are $\frac{n+1}{2}th$ and $\frac{n+3}{2}th$ terms.

- If $n$ is even there will be one middle term in the expansion of $(x+a)^n$ which is $\left(\frac{n}{2}+1\right)th$ term.
- If $n$ is odd there will be two greatest binomial coefficients in the expansion which are ${}^nC_{\frac{n-1}{2}}$ and ${}^nC_{\frac{n+1}{2}}$. Here ${}^nC_{\frac{n-1}{2}}={}^nC_{\frac{n+1}{2}}$.
- If $n$ is even there will be one greatest binomial coefficients in the expansion which is ${}^nC_{\frac{n}{2}}$.
- The coefficient of $x^k$ in the expansion of $\left(ax^p+\frac{b}{x^q}\right)$ is ${}^nC_r a^{n-r} b^r$ where $r=\frac{np-k}{p+q}$.
- The term independent of $x$ in the expansion of $\left(ax^p+\frac{b}{x^q}\right)$ is ${}^nC_r a^{n-r} b^r$ where $r=\frac{np}{p+q}$.
- **<u>Numerically greatest term (NGT)</u>** in the expansion of $(1+x)^n$:
  1. NGT = $(p+1)th$ terms and its value is $|T_{p+1}|$ if $\frac{(n+1)|x|}{|x|+1}=p+f$, where p is an integer and f is a proper fraction and $0<f<1$.
  2. NGT = $(p+1)th$ and $(p)th$ terms and its value is $|T_{p+1}|=|T_p|$ if $\frac{(n+1)|x|}{|x|+1}=p$, where p is an integer.
- To find numerically greatest term of $(a+b)^n$ we have to represent $(a+b)^n=a^n\left(1+\frac{b}{a}\right)^n$ and then take $x=\frac{b}{a}$ and find the term as above.
- The number of non zero terms in the expansion of $(x+a)^n+(x-a)^n$ is $\frac{n+1}{2}$ if $n$ is odd.
- The number of non zero terms in the expansion of $(x+a)^n+(x-a)^n$ is $\frac{n}{2}+1$ if $n$ is even.
- The number of non zero terms in the expansion of $(x+a)^n-(x-a)^n$ is $\frac{n+1}{2}$ if $n$ is odd.
- The number of non zero terms in the expansion of $(x+a)^n-(x-a)^n$ is $\frac{n}{2}$ if $n$ is even.
- The number of terms in the expansion of $(x_1+x_2+...+x_r)^n$ is ${}^{n+r-1}C_{r-1}$.
- The number of terms in the expansion of $(x+y+z)^n$ is $\frac{(n+1)(n+2)}{2}$.
- If $n>2,\ n\in N$, then $(2n-1)^n+(2n)^n<(2n+1)^n$.
- If the coefficients of $x^{r-1}, x^r, x^{r+1}$ are in A.P., then $(n-2r)^2=n+2$.
- $(1+\sim)^n-1$ is divisible by $M(\sim)$.
- $(1+\sim)^n-n\sim-1$ is divisible by $M(\sim^2)$.

- $(1+\sim)^n - {}^nC_2\sim^2 - n\sim -1$ is divisible by $M(\sim^3)$.
- Coefficient of $x^{n-1}$ in $(x-\sim_1)(x-\sim_2)...(x-\sim_n)$ is $-(\sim_1+\sim_2+...+\sim_n)$.
- Coefficient of $x^{n-1}$ in $(x+\sim_1)(x+\sim_2)...(x+\sim_n)$ is $(\sim_1+\sim_2+...+\sim_n)$.
- Coefficient of $x^{n-2}$ in $(x-\sim_1)(x-\sim_2)...(x-\sim_n)$ is -------→
  $\dfrac{(\sim_1+\sim_2+...+\sim_n)^2-(\sim_1{}^2+\sim_2{}^2+...+\sim_n{}^2)}{2}$.
- **<u>Standard Result on Binomial Coefficients:</u>**
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1. $C_0+C_1+C_2+...+C_n=2^n \Rightarrow \sum_{r=0}^{n} C_r = 2^n$.
2. $C_0-C_1+C_2-C_3+...+(-1)^n C_n = 0$.
3. $C_0+C_2+C_4+...=C_1+C_3+C_5+...=2^{n-1}$.
4. $aC_0+(a+d)C_1+(a+2d)C_2+...+(a+nd)C_n=(2a+nd)2^{n-1}$.
5. $aC_0-(a+d)C_1+(a+2d)C_2-...+(-1)^n(a+nd)C_n=0$.
6. $C_1+2C_2+3C_3+...+nC_n=n2^{n-1} \Rightarrow \sum_{r=0}^{n} rC_r = n2^{n-1}$.
7. $C_1-2C_2+3C_3-...+(-1)^{n-1}nC_n=0 \Rightarrow \sum_{r=0}^{n}(-1)^{r-1}rC_r=0$.
8. $C_0+\dfrac{C_1}{2}+\dfrac{C_2}{3}+...+\dfrac{C_n}{n+1}=\dfrac{2^{n+1}-1}{n+1}$.
9. $C_0-\dfrac{C_1}{2}+\dfrac{C_2}{3}-...+(-1)^n\dfrac{C_n}{n+1}=\dfrac{1}{n+1}$.
10. $\dfrac{C_0}{1}+\dfrac{C_2}{3}+\dfrac{C_4}{5}+...=\dfrac{2^n-1}{n+1}$.
11. $\dfrac{C_1}{2}+\dfrac{C_3}{4}+\dfrac{C_5}{6}+...=\dfrac{2^n}{n+1}$.
12. $aC_0{}^2+(a+d)C_1{}^2+(a+2d)C_2{}^2+...+(a+nd)C_n{}^2=\dfrac{(2a+nd)}{2}\left({}^{2n}C_n\right)$.
13. ${}^mC_0\,{}^nC_r+{}^mC_1\,{}^nC_{r-1}+{}^mC_2\,{}^nC_{r-2}+...+{}^mC_r\,{}^nC_0={}^{m+n}C_r$.
14. $C_0+(C_0+C_1)(C_0+C_1+C_2)+...+(C_0+C_1+C_2+...+C_{n-1})=n2^{n-1}$.
15. $C_0{}^2+C_1{}^2+C_2{}^2+...+C_n{}^2={}^{2n}C_n=\dfrac{(2n)!}{(n!)^2}$.
16. $C_0{}^2-C_1{}^2+C_2{}^2-...+(-1)^nC_n{}^2=0$ if $n$ is odd.
17. $C_0{}^2-C_1{}^2+C_2{}^2-...+(-1)^nC_n{}^2=(-1)^{n/2}\times{}^nC_{n/2}$ if $n$ is even.
18. If ${}^nC_x={}^nC_y$, then either $x=y$ or $x+y=n$.
19. ${}^nC_r+{}^nC_{r-1}={}^{n+1}C_r$.
20. $n\,{}^nC_r=(n-r+1)\,{}^nC_{r-1}$.

21. $\dfrac{{}^nC_r}{{}^nC_{r-1}} = \dfrac{n-r+1}{n}$.

- If $f(x) = \left(a_0 + a_1x + a_2x^2 + ... + a_mx^m\right)^n$, then
  1. Sum of the coefficients of x = $f(1) = \left(a_0 + a_1 + a_2 + ... + a_m\right)^n$.
  2. Sum of Coefficients of x having even power $\dfrac{f(1)+f(-1)}{2}$.
  3. Sum of Coefficients of x having odd power $\dfrac{f(1)-f(-1)}{2}$.
- **Multinomial Theorem for a Positive Integral Index**: If $x_1, x_2, ..., x_k$ are real numbers and $n \in N$, $\left(x_1 + x_2 + ... + x_k\right)^n = \sum\limits_{r_1+r_2+...+r_k=n} \dfrac{n!}{r_1! \times r_2! \times ... \times r_k!} x_1^{r_1} x_2^{r_2} ... x_k^{r_k}$ where $r_1, r_2, ..., r_k$ are all non negative integers.

  (a) The general term in the expansion is $\dfrac{n!}{r_1! \times r_2! \times ... \times r_k!} x_1^{r_1} x_2^{r_2} ... x_k^{r_k}$.

  (b) The total number of terms in the expansion = number of non-negative integral solution of $r_1 + r_2 + ... + r_k = n$ is ${}^{n+k-1}C_n$ or ${}^{n+k-1}C_{k-1}$.

  (c) Coefficient of $x_1^{r_1} x_2^{r_2} ... x_k^{r_k}$ in the expansion of $\left(a_1x_1 + a_2x_2 + ... + a_kx_k\right)^n$ is $\dfrac{n!}{r_1! \times r_2! \times ... \times r_k!} a_1^{r_1} a_2^{r_2} ... a_k^{r_k}$.

  (d) Greatest coefficient in the expansion of $\left(x_1 + x_2 + ... + x_k\right)^n$ is $\dfrac{n!}{(q!)^{k-r} \times [(q+1)!]^r}$ where q is the quotient and r is the remainder when n is divided by k.

  (e) Sum of all the coefficient in the expansion of $\left(x_1 + x_2 + ... + x_k\right)^n$ is $k^n$, i.e. by putting each $x_i$ as 1.
- **Important Theorem**:

  If $\left(\sqrt{A} + B\right)^n = I + f$ where I and n are positive integers,

  If n is odd and $0 \le f < 1$, then $(1 + f)f = K^n$ where $A - B^2 = K > 0$ and $\sqrt{A} - B < 1$.

  If n is even integer, then $\left(\sqrt{A} + B\right)^n + \left(\sqrt{A} - B\right)^n = I + f + f'$, so $f + f'$ is an integer. So $f + f' = 1 \Rightarrow f' = 1 - f$.

  Hence $(I + f)(I - f) = (I + f)f' = \left(\sqrt{A} + B\right)^n \left(\sqrt{A} - B\right)^n = \left(A - B^2\right)^n = K^n$.
- **Binomial theorem for rational index**: If $n$ is not a positive integer and $|x| < 1$ then
  1. $(1+x)^n = 1 + nx + \dfrac{n(n-1)}{2!}x^2 + \dfrac{n(n-1)(n-2)}{3!}x^3 + ... + \dfrac{n(n-1)(n-2)...(n-r+1)}{r!}x^r + ..\infty$

     and its general term is $\dfrac{n(n-1)(n-2)...(n-r+1)}{r!}x^r$.

2. $(1-x)^n = 1 - nx + \frac{n(n-1)}{2!}x^2 - \frac{n(n-1)(n-2)}{3!}x^3 + ... + (-1)^r \frac{n(n-1)(n-2)...(n-r+1)}{r!}x^r + ..\infty$

   and its general term is $(-1)^r \frac{n(n-1)(n-2)...(n-r+1)}{r!}x^r$.

3. $(a+x)^n = \left[a\left(1+\frac{x}{a}\right)\right]^n = a^n\left(1+\frac{x}{a}\right)^n = a^n\left[1 + n\frac{x}{a} + \frac{n(n-1)}{2!}\left(\frac{x}{a}\right)^2 + \frac{n(n-1)(n-2)}{3!}\left(\frac{x}{a}\right)^3 + ...\infty\right]$

   and the expansion is valid when $\left|\frac{x}{a}\right| < 1 \Rightarrow |x| < |a|$ and its general term is $\frac{n(n-1)(n-2)...(n-r+1)}{r!}\left(\frac{x}{a}\right)^r a^n$.

4. $(1+x)^{-n} = 1 - nx + \frac{n(n+1)}{2!}x^2 - \frac{n(n+1)(n+2)}{3!}x^3 + ... + (-1)^r \frac{n(n+1)(n+2)...(n+r-1)}{r!}x^r + ..\infty$

   and its general term is $(-1)^r \frac{n(n+1)(n+2)...(n+r-1)}{r!}x^r$.

5. $(1-x)^{-n} = 1 + nx + \frac{n(n+1)}{2!}x^2 + \frac{n(n+1)(n+2)}{3!}x^3 + ... + \frac{n(n+1)(n+2)...(n+r-1)}{r!}x^r + ..\infty$ and its general term is $\frac{n(n+1)(n+2)...(n+r-1)}{r!}x^r$.

6. There are infinite number of terms in the expansion of $(1+x)^n$, where $n$ is a rational index.

7. If x is so small that its square and higher powers may be neglected, then approximate value of $(1+x)^n = 1 + nx$.

8. $(1-x)^{-1} = 1 + x + x^2 + x^3 + ... + x^r + ...\infty$.

9. $(1+x)^{-1} = 1 - x + x^2 - x^3 + ... + (-1)^r x^r + ...\infty$.

10. $(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + ... + (r+1)x^r + ...\infty$.

11. $(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + ... + (-1)^r (r+1)x^r + ...\infty$.

12. $(1-x)^{-3} = \frac{1\times 2 + 2\times 3x + 3\times 4x^2 + 4\times 5x^3 + ... + (r+1)(r+2)x^r + ...\infty}{2}$.

13. $(1+x)^{-3} = \frac{1\times 2 - 2\times 3x + 3\times 4x^2 - 4\times 5x^3 + ... + (-1)^r (r+1)(r+2)x^r + ...\infty}{2}$.

14. $(1-x)^{-\frac{p}{q}} = 1 + \frac{p}{1!}\left(\frac{x}{q}\right) + \frac{p(p+q)}{2!}\left(\frac{x}{q}\right)^2 + ... + \frac{p(p+q)(p+2q)...(p+(r-1)q)}{r!}\left(\frac{x}{q}\right)^2 + ...\infty$

15. $(1+x)^{-\frac{p}{q}} = 1 - \frac{p}{1!}\left(\frac{x}{q}\right) + \frac{p(p+q)}{2!}\left(\frac{x}{q}\right)^2 - ... + (-1)^r \frac{p(p+q)(p+2q)...(p+(r-1)q)}{r!}\left(\frac{x}{q}\right)^2 + ...\infty$

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